

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2022-23

Zoology

Paper: ZOB-403A — Biology: Applied Zoology (Ancillary Course)

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False, giving reasons in 2-3 sentences each: [2 × 6 = 12]

- (i) Lac is mentioned in *Mahabharata*.
- (ii) India doesn't offer a huge potential for aquaculture development.
- (iii) DDT is good for the environment.
- (iv) Spectacled Cobra is found in North-East India.
- (v) *Cirrhinus cirrhosus* is among India's major indigenous carp.
- (vi) *Bombyx mori* produces Tasar silk.

(b) Define the following terms: [1 × 6 = 6]

- (i) Beeswax
- (ii) Mariculture
- (iii) Polyculture
- (iv) Benthic effect
- (v) Eutrophication
- (vi) Estuaries

(c) Differentiate between Krait and Cobra. [2]

Question 2. How is silk produced and what is its economic importance? [10]

Question 3. Discuss the various tasks that worker honeybees perform in the colony. Discuss the way honey is prepared from nectar. [10]

Question 4. How are transgenic animals made? Discuss their importance with one example. [10]

Question 5. Write a detailed note with illustration on the life cycle of the Lac insect. [10]

(Laid in chambers);

ode (nymph) at (3.5,1) [draw, rounded corners, fill=orange!5] **Crawlers (Nymphs)**
(Emergence & search for host);

ode (settle) at (3.5,-1.5) [draw, rounded corners, fill=orange!10] **Settlement**
(Attach to twigs);

ode (lac) at (0,-3.5) [draw, rounded corners, fill=yellow!10] **Lac Secretion**
(Resin cell formation);

ode (adults) at (-3.5,-1.5) [draw, rounded corners, fill=green!10] **Adult Differentiation**
(Wingless Female / Winged Male);

ode (fertilization) at (-3.5,1) [draw, rounded corners, fill=blue!5] **Fertilization**
(Mating / Parthenogenesis);

[->, thick] (egg) – (nymph); [->, thick] (nymph) – (settle); [->, thick] (settle) – (lac); [->, thick] (lac) – (adults); [->, thick] (adults) – (fertilization); [->, thick] (fertilization) – (egg);

- (c) Attraction and repulsion cage mechanism
- (d) Pearl formation

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